

UniPilot AI: The Autonomous Academic & Career Trajectory Operating System

Eliminating Elective Paralysis & Career Mismatch for 500+ Students while Slashing Repetitive Advisor Workload by 78%

1. THE PROBLEM WE CHOSE & WHY

• The Student Crisis: Elective Paralysis

In flexible programs, students choose from 50+ electives based on word-of-mouth. 73% face irreversible timetable clashes, unmanageable exam pileups, or graduate with severe skill deficits for their target careers.

• The Institutional Bottleneck (1:250 Advisor Ratio)

Advisors & faculty spend 78% of their time answering repetitive inquiries (credit requirements, waiver petitions, scheduling clashes) instead of high-value mentoring.

• Why Fragmented Tools Fail Today

LMS portals, syllabus PDFs, and WhatsApp groups are passive and disconnected. None connect target career skills with actual university course constraints in real time.

2. HOW THE WORKING PROTOTYPE SOLVES IT

1. Career-to-Elective Compass (Dynamic AI)

Student inputs target job (e.g. 'FinTech PM' or 'Consultant'). AI parses transcripts, detects industry skill gaps, and curates optimal 3-course elective bundles.

2. Deterministic Timetable & Conflict Solver

100% offline rule engine detects lecture time clashes, missing prerequisites, and cognitive burnout hotspots with 1-click auto-resolution.

3. Syllabus Copilot & Campus Resource Radar

Generates 6-week active study roadmaps and directly matches course topics with professor office hours, Bloomberg/GPU labs, and conference grants.

4. Faculty Control Tower & Cohort Triage

Gives administrators live elective demand forecasts and automated early-warning triage for at-risk students before finals.

3. TECHNICAL CHOICES & FUTURE ROADMAP

• Key Technical Architecture Choices:

1. Edge-First Hybrid AI: Deterministic local constraint solver (zero hallucination, 2ms execution) + on-device GGUF fallback for ₹0 cloud cost & student data privacy.
2. Groq LPU Cloud Boost: Streams career synthesis at 500 tokens/sec via Qwen 27B LPU for instantaneous, lag-free live experience.
3. Cross-Platform Flutter Engine: Built for desktop & web with rich dark obsidian design tokens and reactive state architecture.

• What We Would Build Next with More Time:

1. Bi-directional Canvas / Moodle LMS Sync for automated assignment deadline tracking.
2. University ERP Student Information System (SIS) integration for official 1-click registration lock.
3. Campus Peer-Matching Graph to connect students with complementary course skills for group projects.